Name: ____________________________ Date: ________________ Period: _________

1. A table of values for $f(x)$ is given below (where x is in seconds and $f(x)$ is in feet).

x	2	10
$f(x)$	8	32

Compute the average rate of change of f on the interval $[2, 10]$. Include appropriate units.

$$\text{ARC} = \frac{f(10) - f(2)}{10 - 2} = \frac{\underline{\hspace{3cm}}}{\underline{\hspace{2.5cm}}} = \underline{\hspace{3cm}}$$

Units: ________________________

2. Evaluate $p(x) = -3x^3 + 2x^2 - x + 5$ at $x = -2$. Show each step.

$p(-2) = -3(\underline{\hspace{1cm}})^3 + 2(\underline{\hspace{1cm}})^2 - (\underline{\hspace{1cm}}) + 5 = \underline{\hspace{3.5cm}}$

3. A scientist says: "As the distance between two magnets doubles, the force between them drops to one-fourth of its original value." Circle the model type that best fits this description and explain in one sentence.

A. Linear **B.** Quadratic **C.** Rational (inverse proportion) **D.** Cubic

Explanation: ____